

OBIELF R&D Log – Research and Development Notes

🔗 Project: OBIELF (OBINexus Executable and Linkable Format)

Goal: Define and implement a portable, clean, system-agnostic ELF variant that can be adopted by NASM, developers, and AST-aware compilers.

🧠 Concept Overview

OBIELF is a portable ELF specification that emphasizes universal compatibility, minimal dependencies, and architecture-aware semantic clarity. It's intended for use in low-level systems, NASM compilers, and future bytecode export pipelines (e.g., OBINexus AST-aware stack).

🎯 R&D Objectives

1. Define ELF layout spec extensions

- `.obi.rodata` for policy/metadata sections
- `.obi.axc` for architecture-crossing optimizations
- `.obi.uuid` for traceable builds

2. Create a `nasm -f obielF64` backend target

- Uses standard ELF structure
- Adds OBIELF-specific markers for loaders and validators

3. Design for AST→ELF mapping

- Each IR node → binary section or note segment
- AXC transformations recorded in `.note.axc_history`

4. Ensure ABI independence

- Use raw syscall numbers, no libc
- Avoid glibc/ld-linux loader dependency

5. Cross-platform compilation tests

- Validate `obielF` output on Ubuntu, Alpine, Arch, WSL
 - Future: ARM64, RISC-V backends
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📄 Naming Considerations

- **OBIELF** — Final spec name
- **NXELF** — Short variant for NASM command line
- **OBXELF** — Developer-facing internal name
- **AXC-ELF** — Optional mode for AXC-aware builds

Use Case Examples

```
# Standard NASM build
nasm -f obiel64 main.asm -o main.o
ld -o main main.o
```

```
# Read ELF sections
readelf -a main
objdump -d main
```

```
section .obi.axc
    db 'AXC optimized: true', 0
section .obi.uuid
    db 'build:9ac3-52e7-4e21-b8a2', 0
```

Research Threads (Open)

- Define ELF note segment for policy trust chains
- Consider ELF32/ELF64 hybrids for cross-compatibility
- Embed AST semantic hashes for validation
- Integrate with GOSI Aura cryptographic model

Next Steps

- ☐ Formal spec draft for OBIELF 0.1
- ☐ NASM plugin or patch prototype
- ☐ Sample builds + **obielf-check** validator tool
- ☐ Public GitHub spec for contributions

"Don't just build binaries. Build *truthful* execution." — OBINexus R&D Core Principle

Maintained by OBINexus Research Division – R&D branch on executable formats and next-gen compilation models.